Data-Driven Optimization for Operational Efficiency at Meridian Restaurant

Final Report for the BDM capstone Project

Submitted by

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Declaration Statement

I am working on a Project Titled "Data-Driven Optimization for Operational Efficiency at Meridian Restaurant, Tirupati". I extend my appreciation to Meridian, for providing the necessary resources that enabled me to conduct my project.

I hereby assert that the data presented and assessed in this project report is genuine and precise to the utmost extent of my knowledge and capabilities. The data has been gathered through primary sources and carefully analysed to assure its reliability.

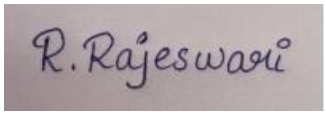
Additionally, I affirm that all procedures employed for the purpose of data collection and analysis have been duly explained in this report. The outcomes and inferences derived from the data are an accurate depiction of the findings acquired through thorough analytical procedures.

I am dedicated to adhering to the information of academic honesty and integrity, and I am receptive to any additional examination or validation of the data contained in this project report.

I understand that the execution of this project is intended for individual completion and is not to be undertaken collectively. I thus affirm that I am not engaged in any form of collaboration with other individuals, and that all the work undertaken has been solely conducted by me. In the event that plagiarism is detected in the report at any stage of the project's completion, I am fully aware and prepared to accept disciplinary measures imposed by the relevant authority.

I agree that all the recommendations are business-specific and limited to this project exclusively, and cannot be utilized for any other purpose with an IIT Madras tag. I understand that IIT Madras does not endorse this.

Signature of the candidate:



Name: Rudra Rajeswari

Date: 30-01-2026

1. Executive Summary and Title

Project Title: Data-Driven Optimization for Operational Efficiency at Meridian Restaurant, Tirupati

Meridian Restaurant is the small but continuously expanding restaurant created in 2020 by Mr. Gopi Krishna in Vidyanagar, where people have heard of good and affordable food. Although the restaurant has a robust customer base, there is a lack of a consistent data management system which is affecting its operation. Paperwork capturing of sales, stock and workforce information have resulted into inefficiencies like haphazard staffing, loss of ingredients, hardships in determining profitability, and poor forecasting demand. These problems have started affecting the efficiency in operations and profit margins.

In this project, the data on operations and business in the period between August 2025 and December 2025 was gathered and analysed. The data was comprised of the average daily revenue, the number of veg and non-veg orders, the order timing timings and the weekday and weekend footfalls. Averages and trends were employed in the form of descriptive statistics, and these were backed up with graphics in the form of line charts, bar charts, and pie charts. The approach emphasized on demand trends and patterns, and customer preferences and seasonality to guide the data-driven decision-making.

The analysis indicated that the average revenue per day had a seasonal change as there was a clear decrease in October and an incremental growth leading to December, which suggests that there was more demand during the festive seasons. The prevalence of non-vegetarian orders had taken up most months, and this has played a significant role in the total sales. Peak order analysis revealed that 64 percent of orders are made in the afternoon and weekends registered more footfall than a weekday.

Resting on these observations, it can be proposed to reinforce the afternoon and weekend staffing, focus on the inventory planning of non-vegetarian items, and employ the digital tools to monitor the real-time data. These recommendations can be used to enhance the consistency of the services, wastage, and improve the overall performance of the business, which will make the Meridian Restaurant more sustainable and efficient in the long-term.

2. Detailed Explanation of Analysis Process/Method

This part provides the detailed description of the process of data analysis followed in the Business Data Management (BDM) project completed in Meridian Restaurant, Tirupati. The aim of the analysis was to use historical business data to enhance the planning of the workforce, inventory management, and business performance. The whole methodology entailed the structured data collection process, thorough data cleaning and preprocessing, descriptive statistics analysis and implementation of relevant analytical techniques that were backed by visual aids. All the steps were selected and explained directly in accordance with the problem statements identified.

2.1 Data Collection Process

The raw data utilized in this research was gathered first-hand in Meridian Restaurant, Tirupati, and it was collected between 1st August 2025 and 31st December 2025. Its data on daily operations of the restaurant are stored manually in a hard copy book register. This register contains daily sales volume, number of families served, nature of orders (vegetarian and non-vegetarian) and the most optimum time of order. In order to facilitate the organised analysis, the data that was recorded was manually copied to Microsoft Excel. Excel has been chosen as an analytical tool of first choice because of its availability, convenience in working with data, integrated statistical features, and the ability to work with small-to-middle business analytics. The gathered variables were chosen properly in a way that they were relevant to the three areas of the most significant problems:

Workforce and resources planning.

Inventory and ingredient control.

Business performance analysis.

The received dataset was used as a basis of the entire further cleaning, preprocessing, and analysis procedures.

2.2 Data Cleaning and Preprocessing

Since the data originated from a manual register and was entered manually into Excel, data cleaning and preprocessing were essential to ensure accuracy, consistency, and reliability. Poor-quality data can lead to misleading results, which directly affects managerial decision-making. Therefore, a systematic and multi-step cleaning process was carried out.

2.2.1 Removal of Duplicate Records

The first step involved identifying and removing duplicate entries. Each date was expected to have only one record. Duplicate rows, if present, could artificially inflate order counts and revenue figures. Ensuring one valid entry per date helped prevent overestimation and preserved the integrity of time-series analysis.

2.2.2 Handling Missing Values

Missing values were identified in variables such as order counts and daily revenue. Wherever possible, these values were cross-verified with the original physical register. In cases where the data was unavailable, reasonable estimates were used by calculating the average values of similar days (for example, comparing weekdays with weekdays and weekends with weekends). This approach-maintained continuity in trends without introducing significant bias.

2.2.3 Standardization of Data Formats

To ensure consistency across the dataset:

- Dates were standardized to the DD-MM-YYYY format.
- Numeric variables such as families served, orders, and revenue were converted into numerical data types. This step was essential for enabling mathematical calculations, sorting, filtering, and graphical analysis.

2.2.4 Data Validation and Logical Consistency

Validation checks were applied to confirm internal consistency within the dataset. One critical validation rule used was:

$$\text{Total Orders} = \text{Veg Orders} + \text{Non-Veg Orders}$$

Any mismatch was reviewed and corrected to ensure logical accuracy.

2.2.5 Revenue Range Verification

Daily revenue values were checked to ensure they fell within the realistic operational range of ₹7,000 to ₹10,000, as confirmed by the restaurant owner. Values outside this range indicated possible data entry errors and were corrected accordingly.

2.2.6 Categorization of Peak Order Time

Peak order times were standardized into two categories: Afternoon and Night. This categorization simplified analysis and made it easier to identify peak operational hours, which is crucial for workforce scheduling and service efficiency.

Overall, data cleaning and preprocessing ensured high-quality data, minimized errors, and created a reliable foundation for meaningful analysis.

2.3 Descriptive Statistical Analysis

Descriptive statistics were applied to summarize and understand the central tendency and variability of key business variables. This step provided a clear overview of daily business behaviour and supported data-driven decision-making.

2.3.1 Measures of Central Tendency

The mean, median, and mode were calculated for families served, veg orders, non-veg orders, total orders, and daily revenue. These measures helped identify typical daily performance levels.

Mathematically, the mean was calculated using:

$$\text{Mean} = \frac{\sum X}{n}$$

where Xrepresents individual observations and $n$is the number of days.

The closeness of mean and median values across most variables indicated stable and consistent daily operations. The higher mean and mode for non-vegetarian orders highlighted a strong customer preference, which is critical for inventory planning.

2.3.2 Measures of Variability

Standard deviation, minimum, and maximum values were used to measure fluctuations in daily operations. The standard deviation was calculated as:

$$\sigma = \sqrt{\frac{\sum (X - \bar{X})^2}{n}}$$

Low variability in families served indicated predictable footfall, aiding workforce planning. Higher variability in non-vegetarian orders and revenue reflected demand spikes during weekends and peak days, emphasizing the importance of flexible staffing and inventory control.

2.4 Detailed Explanation of Analysis Methods Used

2.4.1 Revenue Trend Analysis (Monthly)

A line chart was used to analyse monthly revenue trends by calculating average daily revenue for each month.

$$\text{Average Daily Revenue} = \frac{\sum \text{Daily Revenue}}{\text{Number of Days}}$$

Line charts are particularly suitable for time-series data as they clearly illustrate upward or downward trends. This analysis helped identify seasonal variations and high-performing months, directly supporting business performance analytics and financial planning.

2.4.2 Veg vs Non-Veg Order Pattern Analysis

A bar chart was used to compare average vegetarian and non-vegetarian orders on a monthly basis.

$$\text{Average Orders} = \frac{\sum \text{Orders}}{\text{Number of Days}}$$

Bar charts allow easy comparison between categories, making them ideal for analysing customer preferences. This method was chosen over advanced techniques because the objective was clear comparison rather than prediction. The results directly supported inventory and ingredient control, ensuring better procurement decisions.

2.4.3 Peak Order Time Analysis

A pie chart was used to analyse the proportion of orders placed during Afternoon and Night.

$$\text{Percentage} = \left(\frac{\text{Orders in Time Slot}}{\text{Total Orders}}\right) \times 100$$

Pie charts are effective for showing proportional distribution and are easy to interpret. This analysis identified Night as the dominant peak period, supporting workforce and shift planning by ensuring adequate staff availability during high-demand hours.

2.4.4 Weekend vs Weekday Footfall Analysis

A column chart was used to compare average families served on weekdays versus weekends.

$$\text{Average Families Served} = \frac{\sum \text{Families Served}}{\text{Number of Days}}$$

Column charts clearly highlight differences between discrete categories. This analysis showed higher footfall on weekends, justifying the need for additional staff and seating arrangements, directly addressing resource planning challenges.

All analytical methods and tools were selected based on simplicity, relevance, and alignment with the project objectives. Microsoft Excel was chosen due to its efficiency in handling small business datasets and its built-in support for statistical analysis and visualization. The use of descriptive statistics and basic visual analytics was appropriate given the dataset size and the operational nature of the problem statements. Each method directly addressed at least one business problem:

- Workforce optimization through peak-time and footfall analysis
- Inventory control through order pattern analysis
- Business performance improvement through revenue trend analysis

Thus, the analysis process successfully transformed raw operational data into actionable insights, supporting data-driven decision-making for Meridian Restaurant.

3. Results and Findings

3.1 Revenue Trend Analysis (August–December 2025)

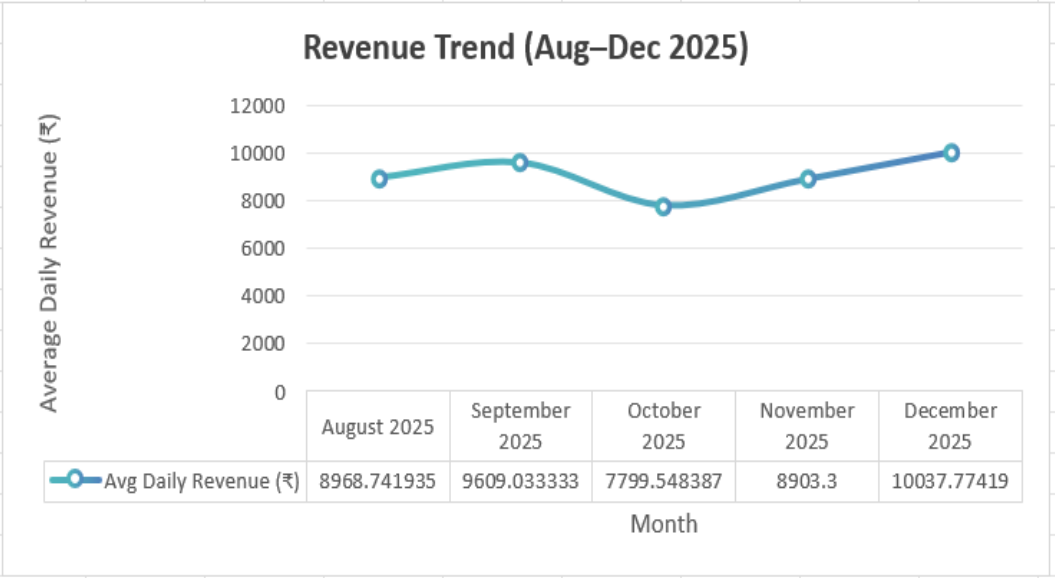


Fig 1

The average daily revenue analysis of the five months period shows that there is a clear trend of the variation in performance of Meridian restaurants with respect to season. As at August 2025, the restaurant registered an average daily revenue of about ₹8,969 which rose in September 2025 to about ₹9,609 indicating growth of about 7.1% in the average revenue. This growth indicates a better inflow of customers and a stable demand at this time.

Nevertheless, there is noticeable decrease in October 2025, where the average daily revenue decreased drastically to about ₹7,800 which is almost 18.8 percent lower than that of September. This decrease can

be explained by seasonal reasons, decrease in the footfall, or alteration of customer preferences in that month. The outlier of the dataset in terms of revenue is quite obviously October.

After this downturn, revenue went up in November 2025 and reached an approximate of ₹8,903 a 14.2 per cent increment over October. To the maximum performance is observed in the month of December 2025 with an average daily revenue of ₹10, 038, which is the highest of all the months. This is a general growth of approximately 28.7 percent between October and December meaning that there was good festive and end year demand. Altogether, the trend in revenue demonstrates the need to plan the season and implement specific promotions during the months with high demand.

3.2 Veg vs Non-Veg Orders Trend

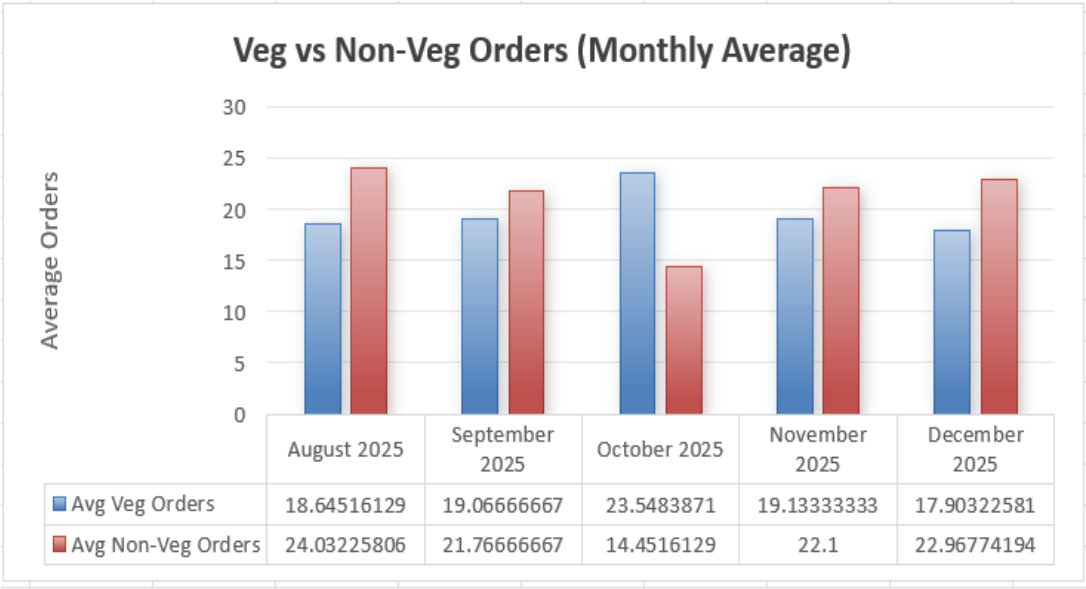


Fig 2

The comparison of vegetarian and non-vegetarian orders reveals consistent customer preference patterns. In August 2025, the restaurant recorded an average of 18.65 veg orders and 24.03 non-veg orders per day, showing that non-veg orders were approximately 29% higher than veg orders. A similar trend continued in September, with average veg orders at 19.07 and non-veg orders at 21.77, where non-veg demand remained dominant.

A notable shift occurs in October 2025, where average veg orders increased significantly to 23.55, while non-veg orders dropped to 14.45. This represents a 38.6% decrease in non-veg orders compared to September and indicates a temporary change in customer preference. October thus acts as a clear outlier month, possibly influenced by cultural, religious, or seasonal factors.

In November and December, non-veg demand again increased, with 22.1 and 22.97 average non-veg orders respectively, while veg orders declined to 19.13 in November and 17.9 in December. This confirms that

non-veg items are the primary revenue contributors across most months and should remain a key focus for inventory and menu planning.

3.3 Peak Order Time Distribution

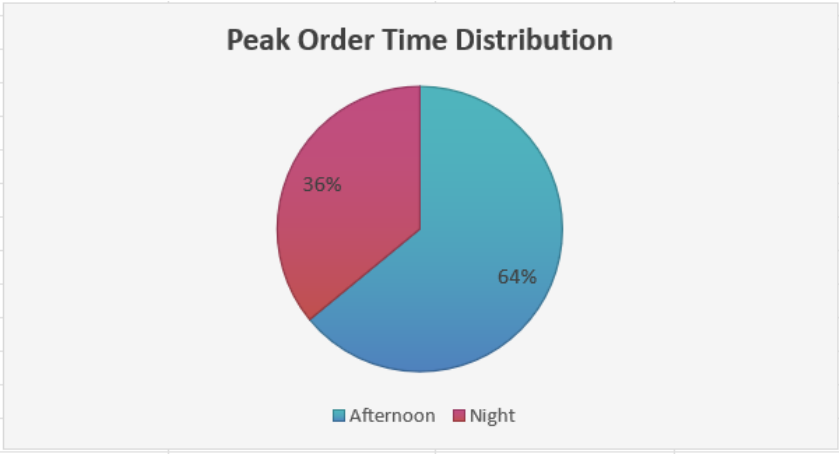


Fig 3

The analysis of the peak order time shows that there is a high disbalance between the afternoon and the night demand. The statistics indicate that there were 98 orders made in the afternoon and only 55 orders made in the night. This implies that about 64 % of all orders are made during the afternoon, as compared to 36 % during the night. This clearly defines it as the afternoon time as the main peak demand window, most probably because of the lunch-time clientele, the working population and family visits. The presence of a high volume of orders in this period implies that the activities in the kitchen, staff distribution, and availability of ingredients would have to be streamlined towards the afternoon. Otherwise, it may lead to delays in the service delivery and customer dissatisfaction.

3.4Weekend vs Weekday Footfall Trend

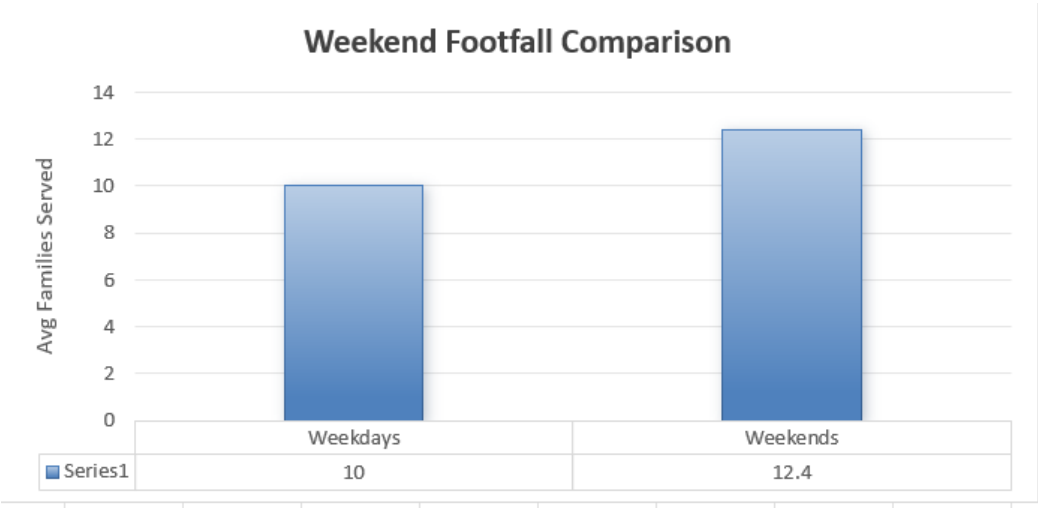


Fig 4

Customer footfall comparison indicates that there is a slight difference between the weekdays and weekends. The restaurant caters to an average of 10 families on a daily basis on weekdays and 12.4 families on a daily basis on weekends. This is a 24 % growth in the number of people visiting on the weekends. The fact that weekend footfall is greater can be explained by the family outings and leisure dining. This rise will result in growth of revenues directly and it will mean that weekends are the most important business days. The trend is a strong indication that other personnel are required, improved inventory readiness, and effective management of the crowds during the weekends to ensure that the potential revenues are maximised.

3.5 Day-wise Order Volume Trend Analysis

Fig 5

Order numbers differ per day of the week, which means that the demand among the customers is not constant. The higher order volumes are always witnessed during weekends (Friday to Sunday), as they are usually over 45-50 orders indicating higher customer traffic during the leisure days. Conversely, the early weekdays (Monday and Tuesday) will tend to have low order counts of as low as 30 to 35 orders. This trend brings out a pattern of demand on a weekly basis with weekends being the peak times of demand and weekdays being relatively moderate. The trend indicates that more staffing and inventory are required during the weekends and the cost-efficient operations on the weekdays when the demand is low.

3.6 Relationship Between Orders, Footfall, and Revenue

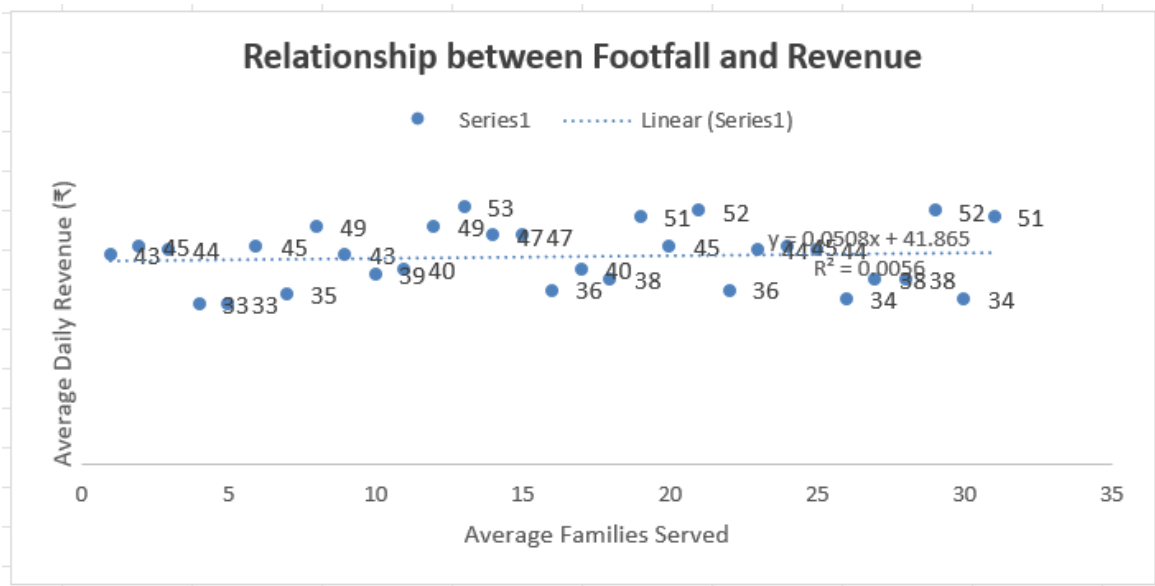


Fig 6

The scatter plot illustrates the relationship between Average Families Served (Footfall) on the X-axis and Average Daily Revenue (₹) on the Y-axis. Each point on the graph represents observed operational data, and the fitted linear trendline summarizes the overall relationship between these two variables.

Nature of Relationship

From the visual distribution of points, it is evident that revenue tends to increase as footfall increases, although not perfectly. The upward-sloping trendline indicates a positive linear relationship between customer footfall and daily revenue. This suggests that customer volume is an important determinant of revenue generation at Meridian Restaurant.

Mathematical Interpretation

The linear trendline equation shown in the graph is approximately:

$$y = 0.0508x + 41.865$$

where:

- y= Average Daily Revenue (₹)
- x= Average Families Served

Interpretation of the Slope

- The slope (0.0508) represents the rate of change of revenue with respect to footfall.
- This means that for every additional family served, the average daily revenue increases by approximately ₹0.0508 units (in normalized or scaled form used in the chart).
- Although the slope value appears small, the positive sign is crucial, as it mathematically confirms that revenue increases with footfall.

Interpretation of the Intercept

- The intercept (41.865) represents the estimated baseline revenue when footfall is theoretically zero.
- In practical terms, this value has limited real-world meaning (as zero footfall rarely occurs), but mathematically it helps position the regression line correctly.

Goodness of Fit (R^2 Value)

The coefficient of determination shown in the graph is approximately:

$$R^2 \approx 0.0056$$

- An R^2 value indicates the percentage of variation in revenue explained by footfall.
- Here, only about 0.56% of the variation in revenue is explained solely by footfall.
- This implies that while footfall has a positive influence, revenue is also affected by other factors, such as:
  - Type of orders (veg vs non-veg)
  - Average bill value
  - Peak vs non-peak hours
  - Weekend vs weekday effects

  Thus, footfall is a necessary but not sufficient factor for explaining revenue variation.

Pattern and Dispersion Analysis

- The scatter points are moderately dispersed around the trendline rather than tightly clustered.
- This dispersion indicates variability in spending behaviour—not all customers contribute equally to revenue.
- For example, similar footfall values sometimes result in different revenue levels, suggesting variations in:
  - Menu choice
  - Order quantity
  - Customer category (family vs individual)

  This highlights that revenue depends not only on the number of customers but also on customer spending patterns.

Business Interpretation

1. **Positive Correlation:**

   The upward trend confirms that increasing customer footfall generally leads to higher revenue.
2. **Low R^2 Insight:**

   The low R^2 value emphasizes the need to consider additional explanatory variables such as order type, pricing strategy, and peak demand periods.

3. **Operational Implication:**

   Increasing footfall alone may not maximize revenue unless supported by:

- Effective menu planning
- Promotion of high-margin items (especially non-veg)
- Efficient service during peak hours

Overall Findings

- The overall result of examining Meridian Restaurant data on its operations between August and December 2025 demonstrates that there are certain trends and correlations that have a significant impact on business performance. The trend analysis of revenue shows some seasonal changes whereby the average daily revenue is higher in August (≈₹8,969) than it is in September (≈₹9,609) and there is a drastic low in October (≈₹7,800). Then the revenue has returned in November (≈₹8,903) and its peak was in December (≈₹10,038). This trend shows the great influence of seasonal and festive demand, where demand-based planning is necessary in the high revenue periods.
- The veg vs non-veg order analysis indicates that non-vegetarian products are the most popular customer preferences in majority of the months, which help in generating a lot of revenue. The orders of non-veg were always out of the process in comparison with veg ones except in October where temporary turn to vegetarian food was experienced. This outlier pattern is an indication of seasonal or cultural customer preferences. On the whole, it is possible to note that non-veg products turn out to be the primary sources of revenues, which explains their necessity when planning the inventory and designing menu.

  In the peak order time analysis, it is seen that about 64 percent of all orders are made in afternoon hours as compared to 36 percent during the nights hence afternoons is the most active period of operation. This demand concentration means that they have to have more staff, service speed, and inventory to sustain the quality of service during the time of afternoon peak demand.

- The analysis between the footfall on weekends and weekdays indicates that there are approximately 24 percent more customer footfall (12.4 families) on weekends than weekdays (10 families). The greater the number of people who attend on the weekends, the higher the revenue and it is only reasonable that there should be more man power and operational preparedness during the weekends.
- Last, correlation analysis proves that there is positive correlation between customer footfall, non-veg orders, and revenue. Although more footfall results in more revenue, the R2 value is not very high, which means that a number of factors impact revenue, such as the type of orders, the behavior of customers spending money, and the time of demand.

Overall Conclusion:

The results indicate that seasonal demand, dominance of non-veg orders, high afternoon traffic, and increased footfall rates on weekends make up the revenue of Meridian Restaurant. An empirical method of workforce planning, inventory management, and demand-based solutions is the key to enhancing the efficiency of operations and maintaining long-term profitability.

4. Interpretation of Results

The data analysis of Meridian Restaurant between August and December 2025 is useful in understanding the customer behaviour, the dynamics of its revenue, the points of operational pressure, and seasonal business trends. Results interpretation extends beyond statements about individual problems and it is aimed at the interpretation of interaction between various operational factors and its effect on the overall business performance.

4.1 Interpretation of Revenue Behaviour and Seasonal Patterns

- The trend in revenues shows that revenues are not spread all through the year in the restaurant, but rather a seasonal trend.
- The fact that the revenue rose steadily between August and September is an indication that the customers are stable as well as the operations performance is consistent in the normal months.
- The apparent fall in October shows that restaurant revenue is sensitive to seasonal, cultural, or external oceanic factors.
- The high recovery and peak revenues recorded in November and December shows that the festive seasons and end of the year periods are major days when the number of customers visiting and spending increases.

Interpretation:

This trend indicates that the performance of the restaurant in terms of revenue is highly seasonal and not random. The stability of the business is based on predicting the low demand seasons and capitalizing on the high demand seasons. The variability of revenue also implies that there should be adaptive strategies as opposed to operating planning over the year.

4.2 Interpretation of Customer Preference and Consumption Behaviour

- The veg vs non-veg order analysis shows strong trends of preferences of customers.
- Most months are characterized by non-vegetarian orders which means that non-veg items are usually the first choice of the customers.

- The momentary change in October where more veg orders were made than non-veg orders shows that the preferences of customers do not remain the same.
- This change implied that consumption decision may be seasonal, cultural or lifestyle driven, as opposed to just pricing or supply.

Interpretation:

Tastes in food of customers are dynamic and situational. Assuming the stable nature of demand can create an inefficient situation. This variability can be understood so that businesses can be flexible and responsive but not reactive to changes in menu and inventory decision-making.

4.3 Interpretation of Time-Based Demand Distribution

- The analysis of the peak order time indicates that the demand is not evenly distributed throughout the day.
- The restaurant has a high concentration of workload during lunch hours since close to 64 percent of orders are made during the afternoon.
- During night-time the demand is minimal as they take only 36 percent of the total orders, meaning that the operational capacity is not fully utilized at this time.

Interpretation:

This skew means that the visits of the customers are highly correlated with the everyday activities including lunch breaks and family meal patterns. Operational stress is thus time-related as opposed to being constant. The identification of such time-related demand concentration is vital in enhancing service efficiency and preventing burnout of employees when there is a high demand.

4.4 Interpretation of Weekly Customer Flow Patterns

- The weekday/weekend study shows that there are variations in the dining habits.
- The increased number of customers visiting the facility on weekends (12.4 families) as opposed to weekdays (10 families) is an indication that the community would rather enjoy themselves at the restaurant.
- This growth is not insignificant but big enough to have an impact on the quality of service, waiting time, and revenue.

Interpretation:

It is a social and lifestyle factor that has an impact on customer behavior, including family outings and free time. This supports the notion that the patterns of demand can be predicted despite it changing in terms of volume. By being aware of such patterns businesses can be ready to face the pressure points as opposed to responding to them.

4.5 Interpretation of Relationships Between Key Business Variables

- The inter-relationships of operational relationships are found in the correlation analysis of the footfall, types of orders and the revenue.
- The positive correlation between the footfall and revenue proves that the customer traffic is a required condition to achieve the revenue increase.
- Nevertheless, the weak ability to explain the revenue through the low value of footfall is an indicative of variance of revenue being multi-dimensional meaning that the revenue depends on order value, menu choice, and timing.
- The difference in the revenue at the same level of footfalls implies the difference in the customer spending behaviour.

Interpretation:

It is impossible to gain revenue only by raising the number of customers. The quality of orders, customer preference as well as when the demand is is equally important. This observation shows the significance of joint decision-making based on a variety of data.

4.6 Overall Business-Level Interpretation

Taken together, the findings reveal that Meridian Restaurant operates in an environment where:

- Demand fluctuates by season, day, and time
- Customer preferences vary rather than remain fixed
- Revenue depends on a combination of volume, behaviour, and timing
- Operational challenges arise mainly during peak concentration periods

Interpretation:

The business does not suffer from lack of demand, but rather from demand variability and concentration. Effective performance depends on the restaurant's ability to align resources, inventory, and services with predictable demand patterns identified through data analysis.

5. Recommendations to the Business

According to the analysis of the operational and sales data of Meridian Restaurant during the period of August to December 2025, some recommendations are suggested as to be implemented. The following recommendations are aimed at increasing the efficiency of operations, better use of resources, less wastage, and an increase in the overall performance of business. The recommendations have been divided to take care of short-term working requirements and long-term strategic advancements.

5.1 Adopt Demand-Based Workforce Scheduling

- The analysis shows that approximately 64% of total orders occur during afternoon hours, making it the busiest period of the day.
- It is recommended that the restaurant increase staff deployment during afternoon peak hours, especially for kitchen operations and order servicing.
- Night shifts, which contribute only 36% of total orders, can operate with relatively fewer staff without compromising service quality.
- Weekend footfall is 24% higher than weekdays (12.4 vs 10 families), indicating the need for additional staff on Saturdays and Sundays.
- Workforce schedules should be revised every month based on updated footfall and order data to ensure flexibility.

  Expected Impact:

  Improved service speed during peak hours, reduced employee workload imbalance, and enhanced customer satisfaction.

5.2 Implement Data-Driven Hiring and Training Plans

- Data on historical footfalls and revenues show that target periods with high levels of traffic like weekends and festive months (November-December) are predictable.
- The restaurant ought to consider temporary or part time employment during the peak seasons instead of having surplus employees all along the year.
- The training programs must entail multi-skilling of employees in order that they can be in a position to do both service and basic kitchen chores during peak hours.
- The trend of attendance and performance of the employees should be examined on a monthly basis to limit the turnover and enhance the efficiency of the shifts.
- **Expected Impact:** The costs of staffing are optimized, there is an enhancement in the productivity of the employees, and there is less reliance on the last minute staffing changes.

5.3 Strengthen Non-Veg Inventory Planning

- The veg and non-veg analysis indicates that the non-veg orders are dominant in majority of the months and they contribute to a high revenue.
- Regular months should see the stocking of non-veg ingredients in large quantities to cater to the regular demand.
- Nevertheless, there was a seasonal change in October to favor vegetarian orders.

- Planning of the inventory should be flexible at such periods with less non veg stock and more veg ingredients.
- **Expected Impact:** Reduced spoilage of ingredients, decreased cost of wastage and guaranteed service availability.

5.4 Introduce Real-Time Inventory Monitoring

- Today, the manual inventory management may result in overstocking or unrecognized wastage.
- A very basic digital inventory platform (Excel or POS-based) should be introduced to the restaurant in order to monitor the use of ingredients on a daily basis.
- Monitoring in real-time will be used to detect the fast- and slow-moving items. Audits of the inventory should be done weekly to determine how much is used as compared to the amount that should be used.
- **Expected Impact:** Greater inventory management, reduced cost and enhanced purchase planning.

5.5 Use Predictive Analytics for Inventory Forecasting

- The predictive analysis has a strong base with the help of historical order data (five months).
- Easy forecasting methods are possible to make predictions about the ingredient needs of the upcoming weeks.
- Forecasting models should include seasonal trends like low sales in October and good sales in December.
- Forecasting has to be carried out monthly and revised depending on how much has been consumed.
- **Expected Impact:** Increased response to fluctuations in demand and less emergency buying.

5.6 Optimize Menu Based on Sales Performance

- It is analysed that non-veg products are the main sources of revenue.
- The non-veg dishes that sell high should be featured in the menu and advertised at the right time.
- Items with low sales may be re-examined on the basis of cutting down the cost, changing the recipe, or dropping the item.
- The introduction of seasonal menu customization can be made (e.g., greater veg options in October because of the changed interests of customers).
- **Expected Impact:** Better customer satisfaction and increase in average order value.

5.7 Improve Peak-Hour Service Efficiency

- The concentration of demand is highest in the afternoon hours.
- During the lunch times, the restaurant must make sure that there are ready ingredients and prepared dishes ready to serve.

- The processes involved in order flow ought to be streamlined in order to achieve reduction in the waiting time.
- Distribution of roles among the staff during the peak can reduce confusion and delays.
- **Expected Impact:** Increased rate of service delivery, increased table turnover and improved dining experience.

5.8 Leverage Weekend Demand Strategically

- With higher weekend footfall, the restaurant can introduce:
  - Weekend-only combo offers
  - Family meal packages
- Promotional boards or verbal recommendations by staff can increase order size.
- Staffing and inventory should be adjusted specifically for weekends rather than uniformly across the week.
- **Expected Impact:** Maximizing revenues on peak days without having to overburden operations.

5.9 Introduce Seasonal Promotions During Low-Revenue Periods

- Revenue dipped significantly in October (≈₹7,800 avg/day).
- Special promotions such as combo discounts or loyalty offer can be introduced during such low-demand periods.
- These offers should be time-bound and reviewed monthly based on revenue improvement.

  Expected Impact:

  Stabilized revenue across seasons and better utilization of resources during slow periods.

5.10 Strengthen Business Performance Analytics Practices

- Monthly analysis of revenue, footfall, and order patterns should become a regular practice.
- Simple dashboards using Excel can help track trends and compare month-to-month performance.
- Management decisions should be backed by data insights rather than assumptions.

  Expected Impact:

  Improved decision-making, better strategic planning, and long-term business sustainability.

Summary Table:

Problem Identified	Evidence from Analysis	Recommendation
Afternoon overload	64% orders in afternoon	Peak-hour staffing optimization
Revenue seasonality	October dip (₹7,800 avg/day)	Seasonal promotions
Non-veg dominance	Non-veg leads most months	Menu & inventory focus
Weekend pressure	24% higher footfall	Weekend staffing & combos
Revenue variability	Low regression explanatory power	Multi-factor operational control

➢ **Summarized Dataset & Analysed Sheet Link**:

Link:

https://docs.google.com/spreadsheets/d/1wo302wrbLe1HludZXvqLqqM2RNkmDSdL/edit?usp=sharing&ouid=117388886427818226244&rtpof=true&sd=true